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Nucleotide Immune Signaling in CBASS, Pycsar, Thoeris, and CRISPR Antiphage Defense

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Keywords

CBASS, Thoeris, type III CRISPR, Pycsar, nucleotide second messenger, antiphage defense

Abstract

Bacteria encode an arsenal of diverse systems that defend against phage infection. A common theme uniting many prevalent antiphage defense systems is the use of specialized nucleotide signals that function as second messengers to activate downstream effector proteins and inhibit viral propagation. In this article, we review the molecular mechanisms controlling nucleotide immune signaling in four major families of antiphage defense systems: CBASS, Pycsar, Thoeris, and type III CRISPR immunity. Analyses of the individual steps connecting phage detection, nucleotide signal synthesis, and downstream effector function reveal shared core principles of signaling and uncover system-specific strategies used to augment immune defense. We compare recently discovered mechanisms used by phages to evade nucleotide immune signaling and highlight convergent strategies that shape host–virus interactions. Finally, we explain how the evolutionary connection between bacterial antiphage defense and eukaryotic antiviral immunity defines fundamental rules that govern nucleotide-based immunity across all kingdoms of life.

Contents

INTRODUCTION	256
CBASS	257
Mechanisms of CD-NTase Activation	258
Role of CD-NTase-Associated Proteins	260
Phage Evasion of CBASS Defense	261
Eukaryotic Conservation	262
PYCSAR	262
Mechanisms of PycC Activation	263
Role of PycC-Associated Proteins	263
Phage Evasion of Pycsar Defense	264
Eukaryotic Conservation	264
THOERIS	264
Mechanism of ThsB Activation	264
Role of ThsB-Associated Proteins	266
Phage Evasion of Thoeris Immunity	267
Eukaryotic Conservation	267
TYPE III CRISPR	267
Mechanisms of Cas10 Activation	268
Role of Type III CRISPR Accessory Proteins	269
Phage Evasion of Type III CRISPR	269
Eukaryotic Conservation	270
CONCLUDING REMARKS AND OPEN QUESTIONS	270

INTRODUCTION

Bacteriophages (phages) constitute the most abundant biological entity on Earth and are likely present in all natural environments that support bacterial life (17). The ubiquity of phages results in constant evolutionary pressure on bacteria to acquire defense systems that can restrict phage infection and replication. The resulting host–virus evolutionary arms race is a major driver of molecular innovation, leading to extensive diversification of host antiphage defense systems and creation of viral mechanisms of immune evasion. Building on well-characterized forms of bacterial antiphage defense, including restriction-modification and CRISPR-Cas systems, recent advances in genome sequencing, bioinformatics, and experimental analysis have enabled the identification of more than 130 novel defense systems (22, 31, 78, 109). This expansion of known antiphage defense systems beyond restriction-modification and CRISPR-Cas systems reveals tremendous diversity in the mechanisms and strategies used by bacteria in the evolutionary battle against phages (9, 46).

Bacterial antiphage defense systems detect phage infection and coordinate an effector response that inhibits viral proliferation. Many systems employ a direct mechanism wherein an individual protein or complex conducts all steps of phage sensing and effector function execution (37). For example, defense-associated sirtuins recognize phage tail tube proteins and degrade the essential metabolite NAD⁺, antiviral STAND homologs detect the highly conserved fold of phage terminase or portal proteins to activate diverse effector domains, and CapRel toxin–antitoxin systems bind phage capsid proteins and pyrophosphorylate transfer RNAs to restrict viral translation (32, 35, 124). In these systems, phage sensing and effector function are physically coupled and typically occur in the same location in the cell. In contrast, a common alternative strategy is a mechanism

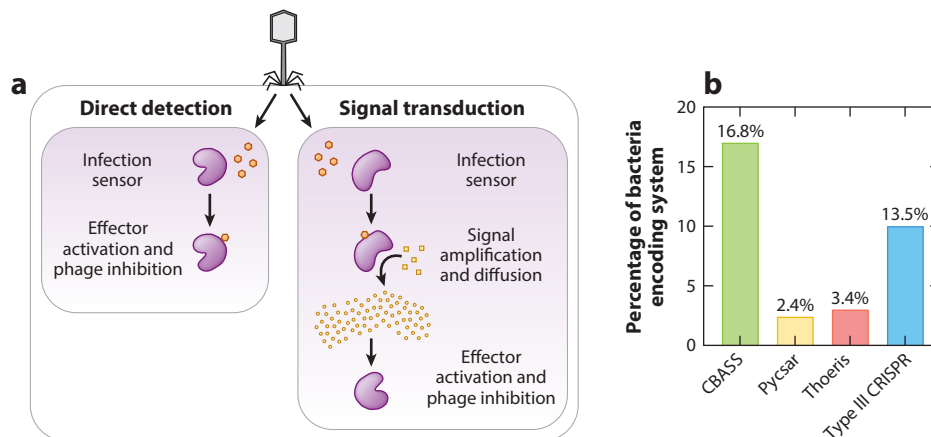


Figure 1

Nucleotide immune signaling pathways provide protection through abortive infection. (a) Bacterial immunity can be generally categorized into direct pathways, which accomplish phage detection and effector function with a single protein or complex, and signal transduction pathways, which use separate components dedicated to detection of incoming phages and execution of effector function. (b) Relative frequencies of major nucleotide immune signaling systems in bacterial antiphage defense. Abbreviation: CBASS, cyclic oligonucleotide-based antiphage signaling system(s). Data are derived from References 106 and 108.

of signal transduction wherein a host protein senses phage infection and then enzymatically synthesizes a diffusible signaling molecule that binds to and activates downstream effector proteins (20, 55, 81, 84, 106) (**Figure 1a**). A distinct advantage of these systems is that functional separation of individual steps of viral recognition and effector execution creates modular systems capable of rapid domain shuffling and diversification to augment host antiphage defense (73).

The most common antiphage defense signaling molecules are specialized nucleotide second messengers. Nucleotides play critical roles in both phage and host replication, and several antiphage defense systems directly modulate the cellular pool of NTPs to interfere with viral replication (see the **Supplemental Material**). In contrast, CBASS, Pycsar, Thoeris, and type III CRISPR antiphage defense systems use the NTP pool to produce diffusible nucleotide immune signals, which enable rapid amplification of pathway signaling responses and allow simultaneous activation of multiple effector proteins throughout the cell. Additionally, nucleotide immune signals in antiphage defense are typically cyclic molecules that lack free phosphate termini and, as a result, are remarkably resistant to degradation and can induce exceptionally potent responses (3, 45). CBASS, Pycsar, Thoeris, or type III CRISPR-Cas occur in ~36% of all bacteria and represent some of the most prevalent forms of antiphage defense (70, 86, 108, 118) (**Figure 1b**). In this review, we focus on recent advances in nucleotide immune signaling in bacteria, with a particular emphasis on molecular mechanisms of immunity, conservation of related pathways in eukaryotes, and immune evasion strategies used by phages.

CBASS

CBASS [cyclic oligonucleotide-based antiphage signaling system(s)] is a set of related antiphage defense systems in bacteria and archaea that were first identified following the discovery of CD-NTase (cGAS/DncV-like nucleotidyltransferase) proteins as structural homologs of an immune signaling enzyme in human cells named cGAS (cyclic GMP-AMP synthase) (20, 114). In CBASS antiphage defense, the CD-NTase senses phage infection and then becomes activated to

Supplemental Material >

enzymatically synthesize a cyclic nucleotide immune signal. CBASS nucleotide immune signals bind to Cap (CD-NTase-associated protein) effectors that induce host cell death to inhibit phage propagation (20, 23, 61, 68) (**Figure 2a**). Minimal type I CBASS operons encode two proteins, the CD-NTase sensor and a Cap effector, and account for ~42% of all currently known systems. The remaining CBASS are classified as types II–IV on the basis of the presence of additional Caps that regulate CD-NTase activity (51, 64, 77, 98, 121). CBASS operons occur in nearly every major bacterial phylum, suggesting that these antiphage defense systems undergo extensive horizontal transfer and can function in diverse cellular environments.

Mechanisms of CD-NTase Activation

CD-NTase proteins adopt a bilobed nucleotidyltransferase (NTase) structure shared with human cGAS and animal cGAS-like receptor proteins (cGLRs) (19, 33, 59, 66, 99, 114, 125, 126). A long N-terminal spine helix braces the NTase core with a C-terminal α -helical lobe and creates an overall cagelike architecture where NTP substrates are coordinated for nucleotide immune signal synthesis (58). Despite a shared overall architecture, CD-NTases exhibit exceptional diversity both in primary amino acid sequence and in the resulting nucleotide immune signal products. Highly variable residues in the lid region of the CD-NTase active site function to template interactions with substrate NTPs and control immune signal specificity (39, 114, 120). Each CD-NTase enzyme synthesizes a single nucleotide immune signal product, but distinct CD-NTases produce a wide array of nucleotide immune signals through synthesis of cyclic di- or trinucleotide species, incorporation of combinations of both purine and pyrimidine bases, and creation of products composed of 3'–5' and/or 2'–5' phosphodiester linkages (68, 114) (**Figure 2b**). CBASS immune systems have been experimentally demonstrated to use more than 10 distinct nucleotide immune signals. At least 180 nucleotide immune signal species are possible in nature, revealing the enormous potential of CD-NTase enzymes to control distinct immune signaling pathways (98).

The mechanisms regulating CD-NTase activation remain incompletely understood. In most cases, CD-NTase enzymes likely use a long cleft formed along the N-terminal spine helix as the primary binding site for recognizing phage-associated molecular patterns (114). In support of this model, mutations to the putative CD-NTase ligand-binding surface disrupt antiphage defense (8, 87), and structurally related cGAS and cGLR enzymes in animal immunity use the same protein surface to recognize activating double-stranded DNA (dsDNA) and dsRNA ligands associated with viral replication (19, 33, 99, 125). A major challenge in verifying this model of CD-NTase activation is that relatively little is known about the identity of the phage molecular cues recognized by CD-NTase enzymes during the initial stage of CBASS activation. The best-characterized example comes from a recent elegant study on staphylococcal phages $\Phi 80a312$ vir and $\Phi NM1\gamma 6$ that allowed mapping of a structured ~400 nt CBASS-activating bacteriophage RNA (cabRNA) recognized by a CdnE CD-NTase enzyme in *Staphylococcus schleiferi* (8). The *S. schleiferi* CD-NTase enzyme binds and responds to phage cabRNAs in vitro, confirming direct recognition of a viral RNA associated with phage infection. Intriguingly, the function of the phage cabRNA remains unknown and many other staphylococcal phages that are restricted by CBASS do not produce cabRNA. CD-NTases are also capable of interacting with phage structural proteins, suggesting that additional mechanisms of CBASS activation remain to be discovered (87).

Structural and functional studies of a CBASS in *Vibrio cholerae* suggest an alternative model of CD-NTase activation. Surprisingly, crystal structures of the CD-NTase DncV (dinucleotide cyclase in *Vibrio*) identified a folate-like molecule bound within the putative primary binding surface (126). Folate-like molecules directly inhibit *V. cholerae* CD-NTase activation in vitro, and folate-depleting antibiotics cause CBASS activation and cell death in vivo (11, 12, 96, 126). On the basis

of these findings, it is likely that some CD-NTase enzymes function as guard immune proteins that indirectly sense phage infection by monitoring levels of key cellular metabolites. In this model, *V. cholerae* DncV is normally in complex with folate-like compounds and held in an inactive state in uninfected cells. Upon phage infection, rapid viral genome replication depletes folate, resulting in ligand dissociation and CD-NTase activation that drives nucleotide immune signal synthesis and CBASS-dependent cell death. *V. cholerae* DncV and the *S. schleiferi* CdnE enzyme that responds to phage *cabRNA* are from extremely divergent positions within the CD-NTase family tree (114), suggesting that many mechanisms of phage recognition may exist throughout diverse CBASS immune systems.

Role of CD-NTase-Associated Proteins

The primary role of CBASS *cap* genes is to function as receptors that recognize the CD-NTase nucleotide immune signal and induce protective cell death (20, 23, 61, 68). Cap receptors exhibit remarkable diversity both in domain architecture and in the molecular mechanisms used to initiate the cell death response (13, 77). These include nuclease domains that degrade host and phage genomic DNA (61, 68), Toll/interleukin-1 receptor/resistance protein (TIR) domains that deplete the essential metabolite NAD⁺ (47, 79), phospholipase domains that cleave membrane lipids (97), and small transmembrane domain-containing proteins that induce nonspecific membrane disruption or form membrane pores (23, 105) (**Figure 2a**). In general, Cap receptors are composed of a discrete nucleotide-binding domain fused to a separate cell death effector domain. Several evolutionarily unrelated domains, including SAVED, STING, and β -barrel domains, are capable of cyclic nucleotide binding, suggesting that this modular architecture enables rapid domain swapping through recombination events (23, 68, 79). In nearly all studied cases, Cap receptor activation occurs through nucleotide immune signal-dependent oligomerization, creating filaments, spirals, or discrete clusters of Cap effectors that rapidly induce effector domain function and cell death. Oligomerization-dependent activation has been observed for diverse CBASS effectors, including Cap4, Cap5, Cap12, Cap15, and NucC, suggesting that oligomerization into a supramolecular complex is a highly effective mechanism to couple nucleotide immune signal recognition with the activities of diverse cell death domains (23, 28, 47, 61, 68, 79, 80).

Beyond the minimal CD-NTase and Cap effector operon architecture that defines type I CBASS operons, type II–IV CBASS operons encode additional Caps that regulate CD-NTase activation. Type II operons are defined by the presence of Cap2, a protein with homology to E1/E2 ubiquitin ligases, and Cap3, a protein with homology to JAB deubiquitinase enzymes. Nearly all CD-NTases in type II operons encode either a glycine or an alanine at the CD-NTase C terminus, which enables Cap2 to covalently attach the CD-NTase to a target molecule (51, 64, 119) (**Figure 2c**). Intriguingly, studies using heterologous expression of diverse type II CBASS operons in *Escherichia coli* illustrated that Cap2 conjugates CD-NTases to a range of cellular proteins, including other CD-NTase protomers, and this conjugation results in increased synthesis of nucleotide immune signals (51, 64, 119). In contrast, a recent study using CBASS operons from *Bacillus* species observed that Cap2 specifically conjugates CD-NTases to the host protein PspA (phage shock protein A), which is proposed to maintain the CD-NTase in an inactive state until phage infection occurs, at which point Cap3 liberates the CD-NTase and allows for its activation (60). In both cases, Cap2 conjugates the CD-NTase to target proteins and is essential for phage restriction, highlighting a critical role for a CD-NTase priming step in type II CBASS immunity. However, the role of Cap3 remains less clear. Cap3 reverses CD-NTase conjugation and opposes Cap2 function, but Cap3 is selectively required in particular combinations of hosts and phages, suggesting a more complex function than simply negative regulation (20, 50, 51, 60, 64).

Type III CBASS operons are defined by the presence of Cap6 and Cap7 accessory factors encoding Trip13 and HORMA domain-containing proteins. In eukaryotes, Trip13–HORMA systems are known to regulate signaling pathways in response to short peptides, and a similar model of regulation has been proposed for type III CBASS (121). HORMA domains function by binding short peptides and undergoing a conformational change to engage target proteins. Trip13 domain-containing proteins act as a negative regulator of HORMA domains by disrupting HORMA–peptide complexes and inducing the HORMA domain into an open, unbound state (40). Structural analyses of Cap6 and Cap7 in type III CBASS immunity demonstrate that Cap7–peptide complexes contribute to CD-NTase activation and that the Cap6 Trip13 domain maintains the Cap7 HORMA domain in an unbound state (61, 121). Similar to what is observed in type II CBASS, these results support a model in which Cap6 and Cap7 have opposing functions in regulating CD-NTase activation and suggest that an unknown peptide produced during phage infection may serve as the molecular cue activating type III CBASS immunity (**Figure 2c**).

Phage Evasion of CBASS Defense

The extreme diversity of mechanisms used by CBASS immunity is likely the result of an evolutionary arms race with phages. Phages are known to encode two distinct classes of anti-CBASS (Acb) proteins, named Acb1 and Acb2 (45, 50, 51) following a nomenclature developed for phage anti-CRISPR (Acr) proteins (11). Acb1 is a nuclease that specifically degrades CBASS nucleotide immune signals, while Acb2 acts as a molecular sponge that sequesters nucleotide signals to prevent effector activation (**Figure 2d**). The mechanisms underlying Acb1 and Acb2 immune evasion provide insights into the evolutionary pressures that contribute to CBASS diversification. For instance, both Acb1 and Acb2 target the nucleotide immune signal rather than CBASS protein components, suggesting that the production of a diffusible signal is a point of vulnerability in antiphage defense. Despite effectively neutralizing 3'3'-cGAMP, both Acb1 and Acb2 have evolved to not target the chemically related host signal 3'3'-c-di-GMP, suggesting that disruption of host 3'3'-c-di-GMP signaling is detrimental to phage replication. A structural and biochemical analysis of phage Acb1 proteins explains how the viral nuclease degrades a wide range of CBASS cyclic di- and trinucleotide immune signals (45). However, Acb1 only weakly targets the CBASS nucleotide immune signals 3'2'-cGAMP and 3'3'-c-di-UMP, suggesting that the nucleobase and phosphodiester linkage diversity observed in CBASS signaling may have resulted from evolution to escape phage nuclease evasion mechanisms. Acb2 homologs from phages with different host ranges bind distinct combinations of CBASS signals, possibly reflecting a preference for signals commonly used by each host (15). Likewise, some phage sponge proteins that target nucleotide immune signals in *Thoeris* antiphage defense are also capable of sequestering cyclic dinucleotides to evade CBASS, demonstrating that individual sponge proteins can target multiple classes of host immunity (65).

Remarkably, eukaryotic viruses and phages have converged on a common strategy to evade host nucleotide immune signaling. Mammalian and insect poxviruses encode a protein named poxvirus immune nuclease (poxin) that cleaves the cGAS product 2'3'-cGAMP to evade host immunity (24). Despite having evolutionarily unrelated folds, Acb1 and poxin use highly similar metal-independent biochemical mechanisms to cleave nucleotide immune signals including 3'3'-cGAMP and 2'3'-cGAMP (24, 45). It is likely that phages encode additional proteins dedicated to evasion of CBASS immunity. Understanding the function of these proteins will help explain conserved immune evasion mechanisms potentially used by viruses to evade cGAS- and cGLR-dependent immune signaling in animal innate immunity.

Eukaryotic Conservation

Early experiments in CBASS immunity led to the discovery of a direct evolutionary connection between bacterial antiphage defense and animal antiviral immunity (20, 79, 114). The identification of CD-NTases as cGAS-like signaling enzymes and Cap effectors with STING domains demonstrated that the core components of human cGAS–STING immunity originated billions of years ago in bacteria as part of defense systems against phage infection (79, 114) (**Figure 2e**). Mechanistic analyses of CBASS continue to provide a foundation to explain regulation of immune signaling in animal cells, including a diverse repertoire of nucleotide signals later mirrored in animal cGLR immune proteins (14, 66, 99) and the mechanism of STING oligomerization into signaling filaments (80). Beyond cGAS and STING, many other Cap genes associated with type II–IV CBASS share homology with animal proteins, including E1/E2 ubiquitin ligases (Cap2) and deubiquitinases (Cap3), Trp13 AAA⁺ ATPases (Cap6) and HORMA domain proteins (Cap7), and TIR enzymes (47, 51, 64, 79, 119, 121). CBASS operons are incredibly diverse, with many protein domains participating in immunity through currently unknown mechanisms (13, 77). Given the high degree of conservation between CBASS and cGAS–STING, studies aiming to elucidate the mechanisms and structures of rare CBASS components are likely to uncover additional connections to eukaryotic antiviral immunity.

PYCSAR

Pycsar is a system of antiphage defense defined by the presence of a PycC (pyrimidine cyclase) enzyme encoded alongside a cNMP (cyclic nucleotide monophosphate)-responsive effector protein (106). Following phage infection, PycC becomes activated and synthesizes a cyclic pyrimidine mononucleotide signal that binds to the associated effector protein to trigger rapid cell death through the depletion of NAD⁺ or the disruption of the bacterial membrane (**Figure 3a**). Pycsar immunity is relatively widespread, with 967 Pycsar operons that have been identified

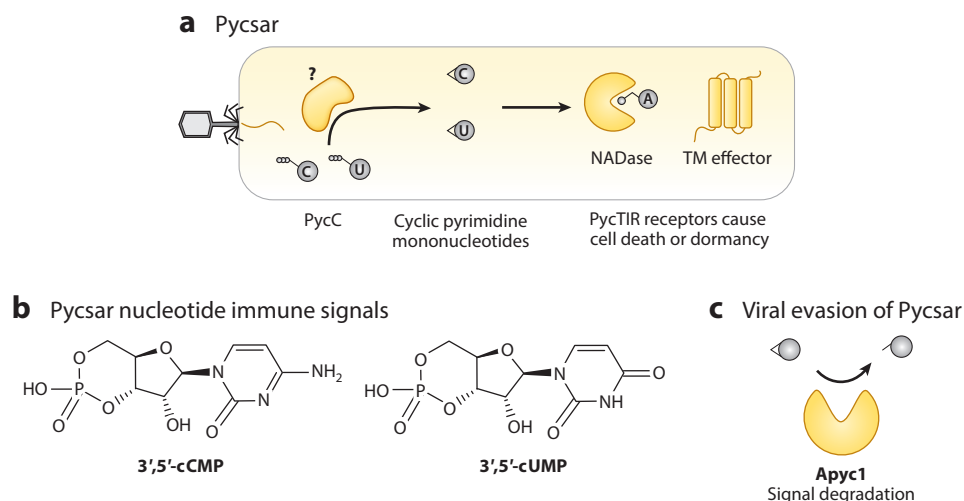


Figure 3

Mechanisms of Pycsar nucleotide immune signaling and phage evasion. (a) Overview of the mechanisms of Pycsar immunity. (b) Chemical structures of the Pycsar signaling molecules 3',5'-cCMP and 3',5'-cUMP. (c) Phages evade Pycsar immunity by targeting nucleotide immune signals with Apyc1. Abbreviations: TIR, Toll/interleukin-1 receptor/resistance protein; TM, transmembrane.

by sequence homology, covering ~2.4% of sequenced bacteria across a diverse range of phyla (106).

Mechanisms of PycC Activation

Biochemical and cellular analyses demonstrated that the PycC proteins, originally annotated as adenylate cyclases, preferentially synthesize cyclic pyrimidine mononucleotide signals including 3',5'-cyclic cytosine monophosphate (cCMP) and 3',5'-cyclic uridine monophosphate (cUMP) (106) (**Figure 3b**). Structures of PycC from *Burkholderia cepacia* and *Pseudomonas aeruginosa* demonstrate that PycC enzymes are dimeric, existing as a homodimer in solution or encoded as two consecutive nucleotide cyclase domains that fold as a single chain into a pseudoheterodimeric structure (67, 106). The PycC dimer forms a composite active site wherein one protomer provides residues for metal coordination and the opposite protomer provides the nucleobase-binding pocket. PycC enzymes are most similar to class III adenylate and guanylate cyclase proteins but lack the canonical residues necessary for coordinating purine nucleobases (67, 102, 106). Instead, PycC proteins encode bulky tyrosine, phenylalanine, and arginine residues in the nucleotide-binding pocket that sterically exclude purine bases in favor of smaller cytosine or uridine pyrimidine substrates (106). Selectivity for pyrimidine bases insulates Pycsars antiphage defense systems from homeostatic cAMP/cGMP signaling and confers a dedicated role for cCMP and cUMP in immune defense (95).

The precise nature of the phage-associated molecular pattern that activates PycC is not understood. Upon infection with phage T5, *E. coli* cells expressing Pycsars exhibit increased cCMP/cUMP levels starting ~30–40 min post infection, suggesting that PycC responds to a molecular cue that occurs at a late stage of viral replication (106). Phage T5 mutants that escape Pycsars immunity acquire spontaneous mutations in capsid proteins, suggesting a possible role for structural proteins expressed late during infection in activating PycC (106). Response to late structural genes is implicated in multiple other abortive infection systems that also cause host cell death (32, 35, 50, 87, 104, 124), raising the intriguing possibility that these potent host defense pathways are reserved as a last resort response and activated only in the final stages of phage replication and capsid assembly. PycC proteins are frequently fused to a small domain named AGS-C, and deletion of this domain from an *E. coli* Pycsars operon ablates antiphage defense against phages T5 and P1. The crystal structure of a Pycsars AGS-C reveals that this domain is closely related to animal immunoglobulin domains (87). AGS-C domains are also sometimes fused to CBASS CD-NTase proteins, supporting a possible role for AGS-C fusions in modulating immune recognition of phage-associated molecular patterns across multiple defense systems (13, 87, 106).

Role of PycC-Associated Proteins

In addition to PycC, Pycsars operons contain *pap* (*Pycsars-associated protein*) genes that recognize cCMP/cUMP signals and induce protective cell death (106). The two main effector types present in Pycsars are PycTIR proteins, which rapidly cleave NAD⁺, and PycTM proteins, which likely use small transmembrane segments to disrupt the bacterial inner membrane. In many cases, Pycsars effector proteins encode an N-terminal domain with predicted homology to cyclic nucleotide-binding domains that respond to cAMP and cGMP (38), indicating that Pap effectors likely use a dedicated domain to recognize activating cCMP and cUMP nucleotide immune signals. Similar to CBASS immunity, cCMP/cUMP-bound Pap effectors oligomerize into large supramolecular complexes to enable activation of the cell death effector domain, illustrating a shared principle of effector regulation across systems (106).

Pycsar operons exist in type I and type II configurations and also encode accessory proteins observed in type II CBASS. In type II Pycsar, Pap2 and Pap3 are direct homologs of CBASS Cap2 (E1/E2-like) and Cap3 (Jab deubiquitinase) proteins that control CD-NTase conjugation to target molecules (51, 64, 119). PycC proteins in type II Pycsar also contain a glycine or alanine residue at the C terminus, suggesting that a similar E1/E2-like conjugation and priming step may be used to regulate cCMP/cUMP signaling, but this mechanism has not been experimentally confirmed (64, 106).

Phage Evasion of Pycsar Defense

Pycsar operons do not typically contain phosphodiesterases that cleave cyclic pyrimidine signals, and it is unclear whether host-encoded phosphodiesterases negatively regulate Pycsar signaling. However, various families of *Bacillus* phages encode anti-Pycsar (Apyc) phosphodiesterases that rapidly cleave Pycsar signals to subvert Pycsar immunity (45). Similar to viral Acb1 and poxin enzymes that target cyclic oligonucleotides, the *Bacillus* phage SBSΦJ protein Apyc1 degrades cCMP and cUMP and inactivates Pycsar defense, highlighting the susceptibility of nucleotide signaling molecules in second messenger-based immune pathways (24, 45) (**Figure 3c**). Sequence alignments of Apyc1 with host-encoded phosphodiesterases that preferentially cleave cAMP and cGMP revealed a two-residue insertion in the Apyc1 active site that may enable base-specific contacts with smaller cyclic pyrimidine signal. Many bacterial genomes encode homologs of Apyc1 in operons distinct from Pycsar antiphage defense. It is not clear if these Apyc1 homologs are located within prophages that enable evasion of Pycsar immunity, or whether they regulate other currently unidentified cCMP- or cUMP-based signaling systems.

Eukaryotic Conservation

Many of the protein components of Pycsar signaling are also found in eukaryotic cells. Pycsar was recently identified as one of the antiviral defense systems specifically enriched in Asgardarchaeota genomes compared with those of other prokaryotes, suggesting a possible evolutionary route for transfer of Pycsar components into early animal lineages (62). Human cells encode multiple enzymes capable of synthesizing cCMP and/or cUMP, and multiple phosphodiesterase enzymes can degrade cCMP, suggesting that some human cells may encode all the necessary components of a complete signaling pathway (7, 94). Although the physiological processes regulated by cCMP or cUMP in eukaryotes remain unclear, cCMP and cUMP have been identified in mammalian tissue extracts using highly sensitive mass spectrometry, further suggesting that cyclic pyrimidine signaling may be conserved across both prokaryotic and eukaryotic domains of life (95, 117).

THOERIS

Thoeris antiphage defense operons are generally composed of two genes, *thsA* and *thsB* (22). During phage infection, ThsB functions as a sensor protein and becomes activated to synthesize cyclic and linear ADP-ribose isomers that bind to associated ThsA effector proteins and inhibit viral replication through the depletion of NAD⁺ or the disruption of the bacterial membrane (22, 63, 72, 91) (**Figure 4a**). Thoeris is distributed broadly across major bacterial phyla and is encoded in ~3.4% of all sequenced bacteria (108).

Mechanism of ThsB Activation

A defining feature of Thoeris antiphage defense pathways is a TIR domain-containing protein named ThsB, which senses phage infection and generates a nucleotide immune signal from the

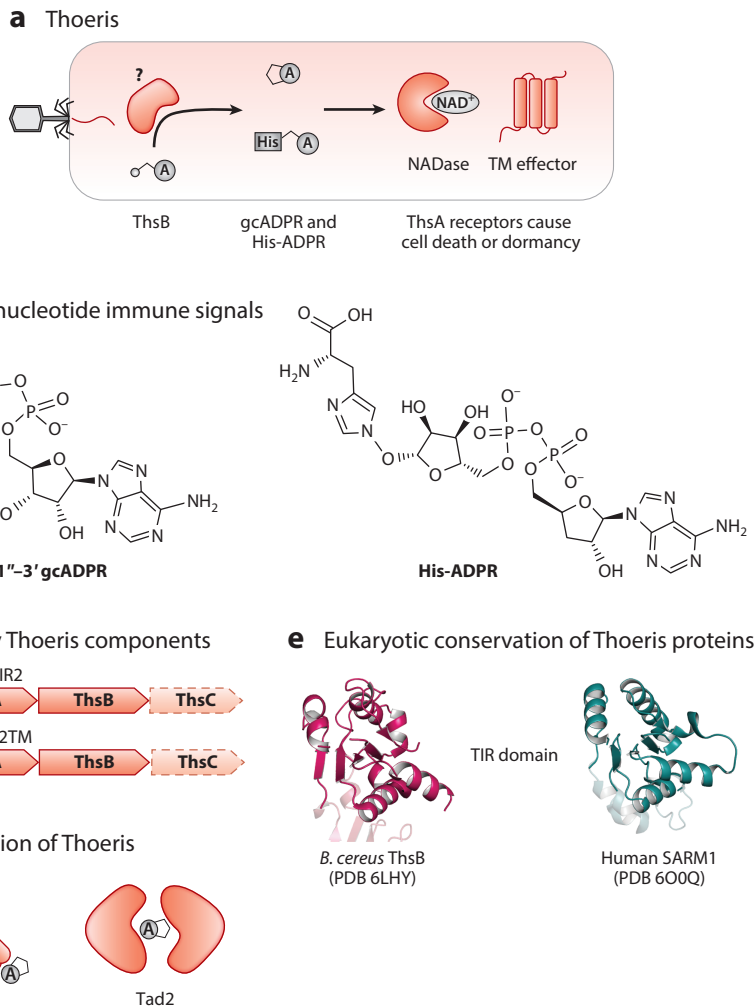


Figure 4

Mechanisms of Thoeis immune signaling and phage evasion. (a) Overview of the mechanisms of Thoeis immunity. (b) Chemical structures of the Thoeis signaling molecules 1''-3' gcADPR and His-ADPR. (c) Thoeis operons contain at least one copy of ThsB and a single copy of ThsA. (d) Phages evade Thoeis immunity by directly targeting nucleotide immune signals. (e) Cartoon representation of the TIR domain structures of *Bacillus cereus* ThsB and human SARM1, highlighting conserved structural features. Abbreviations: gcADPR, glycocyclic ADP-ribose; His-ADPR, histidine ADP-ribose; PDB, Protein Data Bank identifier; SARM1, sterile α and TIR motif-containing protein 1; TIR, Toll/interleukin-1 receptor/resistance protein; TM, transmembrane.

cellular metabolite NAD⁺. TIRs are small, ~150-amino-acid domains that adopt a so-called Rossmann fold consisting of alternating β -strands and α -helices that form a nucleotide-binding pocket to coordinate NAD⁺. TIRs typically have one of three known functions: (a) TIR enzymes that use NAD⁺ as a substrate to create nucleotide immune signals (e.g., Thoeis ThsB and plant NLR proteins) (49, 63, 72, 84), (b) TIR enzymes that are NADases and rapidly cleave NAD⁺ as part of an effector function (e.g., CBASS Cap12 and Pycsar PycTIR) (26, 48, 79, 84, 106, 111), or (c) catalytically inactive TIRs that appear to have lost the capacity to bind NAD⁺ and instead

mediate protein–protein interactions to control signal transduction (e.g., human Toll-like receptors and MyD88) (30, 110). ThsB converts NAD^+ into a nucleotide immune signal by using a highly conserved glutamate residue within the TIR active site to catalyze the release of nicotinamide from ADP-ribose, leaving the 1'' position of ADP-ribose available for nucleophilic substitution (3, 63, 72, 84). In type I Thoeis systems, ThsB produces the nucleotide immune signal 1''–3' gcADPR (glycocyclic ADP-ribose), a cyclic ADP-ribose isomer that contains a 1''–3' ribose–ribose bond (63, 72) (**Figure 4b**). In type II Thoeis, ThsB conjugates NAD^+ to the metabolite histidine to create the linear nucleotide immune signal His-ADPR (91). Similar to CBASS immunity, it is likely that ThsB proteins can synthesize an array of NAD-based nucleotide products to diversify host antiphage defense signaling.

The molecular mechanism controlling ThsB activation in response to phage infection is poorly understood. Experiments with chimeric Thoeis operons confirm that ThsB is responsible for phage-associated molecular pattern recognition and is alone sufficient to control immune specificity against distinct phages (84). However, ThsB proteins exhibit enormous sequence diversity, and there are no apparent conserved surfaces on the structure of the ThsB TIR that are known to mediate phage-associated molecular pattern recognition (54). Although the phage-associated activating cue is currently unknown, gcADPR production during type I Thoeis signaling occurs only at late time points during the phage infection cycle, suggesting that recognition of a late-stage replication component is required for ThsB activation (63, 84). All known enzymatic TIRs become activated through protein oligomerization and formation of supramolecular complexes that can take the form of filaments, clusters, or lattice sheets (47, 48, 79, 80, 106). ThsB is monomeric in solution (54), suggesting that recognition of phage-associated molecular patterns likely induces ThsB self-association and oligomerization to drive activation and nucleotide immune signal synthesis.

Role of ThsB-Associated Proteins

ThsA effector proteins respond to the ThsB-produced nucleotide signal and induce cell death or cell dormancy to inhibit phage propagation. In type I Thoeis, ThsA proteins most often exhibit a two-domain architecture wherein a nucleotide-binding SLOG domain recognizes 1''–3' gcADPR and an appended SIR2 NADase domain depletes cellular NAD^+ and halts viral replication (22, 63, 72, 84). Following 1''–3' gcADPR binding, type I ThsA proteins undergo oligomerization and conformational changes that reposition the active-site residues of the SIR2 domain and allow rapid NAD^+ degradation (54, 72, 84). A peculiar facet of type I Thoeis is the use of NAD^+ both by the ThsB TIR domain, as a precursor to generate a nucleotide immune signal, and by the ThsA SIR2 domain, as the target of NADase effector function. Depletion of NAD^+ by ThsA may serve as a negative-feedback loop to limit ThsB 1''–3' gcADPR synthesis and allow recovery from cell dormancy (63).

In type II Thoeis systems, the ThsA effector protein is usually composed of small transmembrane helices fused to a C-terminal Macro nucleotide-binding domain (84, 91) (**Figure 4c**). The Macro domain recognizes the nucleotide immune signal His-ADPR and likely controls activation through protein oligomerization. The mechanism of type II ThsA cell death is unknown, but it likely occurs through inner membrane disruption similar to transmembrane-containing CBASS Cap effectors (23, 91).

Thoeis operons exhibit additional varied configurations; approximately half of all Thoeis operons encode multiple ThsB copies adjacent to a single copy of a ThsA effector, which may allow cells to respond to distinct infection cues that converge on activation of ThsA (84). In addition to the core components of ThsA and ThsB, ~2.4% of Thoeis operons contain a third gene, *thsC*,

which encodes a histidine triad domain-containing protein that may function as a hydrolase to regulate turnover of ThsB-produced nucleotide immune signals (86) (**Figure 4c**). The unknown function of these accessory proteins highlights an important gap in our understanding of Thoeis immunity.

Phage Evasion of Thoeis Immunity

Phages encode Tad (Thoeis antidefense) proteins that enable evasion of Thoeis immunity. First identified through comparison of the genomes of closely related *Bacillus* phages that exhibit varying resistance to Thoeis defense, Tad1 and Tad2 are founding members of two families of phage proteins that function as molecular sponges to tightly bind nucleotide immune signals and prevent ThsA activation (63, 122) (**Figure 4d**). Diverse phage Tad1 and Tad2 homologs exhibit distinct specificities for 1''–3' gcADPR and His-ADPR nucleotide immune signals, suggesting an extensive arms race between host diversification of Thoeis signaling and viral immune evasion (63, 65, 91, 122). Recent bioinformatic surveys have found that phages encode a vast array of protein domains that degrade, bind, or synthesize NAD⁺-like molecules (3), suggesting that there is a tremendous diversity of Thoeis evasion mechanisms remaining to be discovered.

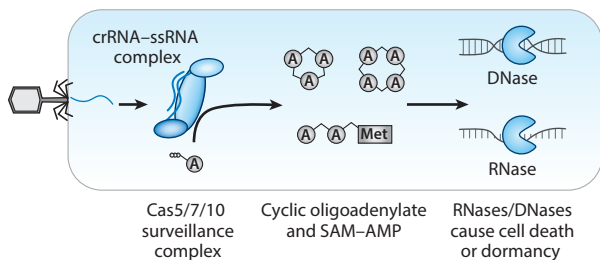
Eukaryotic Conservation

TIR domain-containing proteins are encoded ubiquitously in bacterial, plant, and animal genomes. Plant NLR proteins respond to pathogen infection and use TIR domains to synthesize nucleotide immune signals, including an isomer of gcADPR with a 1''–2' ribose–ribose linkage (63, 72), and linear ADPR signals, including pRib-AMP [2'-(5''-phosphoribosyl)-5'-adenosine monophosphate] (49). A human TIR domain-containing protein involved in axonal degradation named SARM1 (sterile α and TIR motif-containing protein 1) exhibits structural homology to ThsB (**Figure 4e**). Human and *Drosophila* SARM1 exhibit robust NADase activity (26) and can produce small numbers of 1''–3' gcADPR and 1''–2' gcADPR molecules in vitro (34). Animal cells also encode proteins with nucleotide-binding Macro domains and ThsA-like domains, suggesting a possible role for gcADPR signaling in animal cells (21, 34).

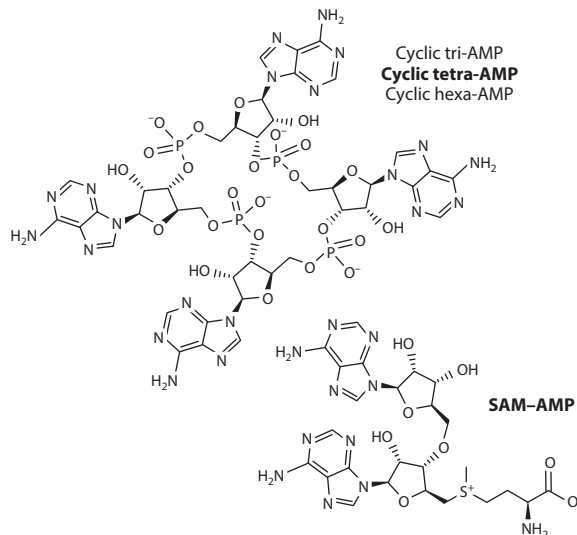
TYPE III CRISPR

CRISPR-Cas systems are an abundant and remarkably diverse family of antiphage defense systems that use RNA guides to target invading nucleic acid (83, 112). First, an acquisition step controlled by Cas1 and Cas2 enzymes converts foreign nucleic acid into spacer segments that are integrated into genomic CRISPR (clustered regularly interspaced short palindromic repeat) loci (74, 82). Next, spacer segments are transcribed into CRISPR RNA (crRNA) guides that, together with Cas (CRISPR-associated) proteins, form ribonucleoprotein complexes that recognize complementary invading nucleic acid and typically catalyze DNA or RNA cleavage to restrict phage replication (112). Approximately half of all sequenced bacteria encode a CRISPR system (37), and these CRISPR-Cas systems use an incredible diversity of molecular mechanisms to couple crRNA-mediated target recognition with effector functions that inhibit phages. Excellent recent reviews (71, 83, 112) present a general overview of the mechanisms of CRISPR antiphage defense. Here, we focus on a common form of antiphage defense named type III CRISPR immunity, which consists of CRISPR-Cas systems that function through production of a nucleotide immune signal (55, 81). In type III CRISPR immunity, a specialized enzyme named Cas10 synthesizes the nucleotide immune signal that then activates downstream effector proteins to induce cell dormancy or cell death (57, 116) (**Figure 5a**).

a Type III CRISPR



b Type III CRISPR nucleotide immune signals



c Viral evasion of type III CRISPR

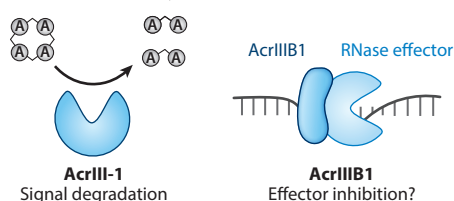


Figure 5

Mechanisms of type III CRISPR nucleotide immune signaling and phage evasion. (a) Overview of the mechanisms of type III CRISPR immunity. (b) Chemical structures of cAAAA and SAM-AMP as representative type III CRISPR nucleotide immune signals. (c) Phages evade type III CRISPR immunity using diverse anti-CRISPR proteins. Abbreviation: SAM, S-adenosyl-methionine.

Mechanisms of Cas10 Activation

A defining feature of type III CRISPR systems is the protein Cas10. Cas10 is a multidomain protein typically composed of an N-terminal HD nuclease domain, two palm domains that control enzymatic activity, a zinc-finger domain, and a C-terminal domain that serves as a binding site for the auxiliary proteins, including Cas5, Cas7, and Cas11 (52, 116, 123). Cas10, auxiliary Cas proteins, and a crRNA guide form a ribonucleoprotein complex that surveys the cell for invading nucleic acid. In contrast to most CRISPR-Cas systems that recognize invading DNA, type III CRISPR complexes typically target single-stranded RNA (ssRNA), including viral mRNA. Target RNA recognition induces a conformational change in the ribonucleoprotein complex that activates Cas10 to produce a cyclic oligoadenylate (cOA) nucleotide immune signal that controls downstream immune defense (55, 81). The two palm domains in Cas10 form a composite active site containing a single GGDD catalytic motif that uses ATP substrates to form the cOA signaling product (52, 123). Although the overall active-site architecture is conserved across diverse Cas10 palm domains, the size of the active site varies, and Cas10 orthologs can synthesize distinct major cOA signaling products ranging between three and six adenosine nucleotides (6, 55, 81) (**Figure 5b**). Highlighting the amplification achieved by nucleotide second messenger systems, biochemical and cellular analyses of a type III CRISPR system from *Sulfolobus solfataricus* demonstrate that Cas10 is capable of synthesizing more than 1,000 cOA molecules for each target RNA binding event (2). In addition to cOA nucleotide immune signals, a recent study showed that a subtype of Cas10 enzymes synthesizes a previously unknown product, SAM-AMP, that comprises S-adenosyl-methionine (SAM) conjugated to AMP (18) (**Figure 5b**). This observation mirrors the recent demonstration of His-ADPR functioning as a type II Thoeris nucleotide immune signal (91) and demonstrates that bacterial antiphage defense systems can broadly include common amino acid and cofactor metabolites to further extend the diversity of host nucleotide immune signals

used to defend against phage infection. The enormous diversity of Cas10 proteins in type III CRISPR systems suggests that additional nucleotide immune signals are likely to be discovered.

In addition to the main role of Cas10 in nucleotide immune signal synthesis, type III CRISPR surveillance complexes can have accessory enzymatic functions that further augment and regulate immune defense. In some type III CRISPR systems, the Cas10 HD domain cleaves ssDNA targets (25, 27, 56). The type III CRISPR Cas7 subunit can also cleave ssRNA that is complementary to the crRNA guide, allowing direct degradation of invading nucleic acid (42, 101). In some type III CRISPR systems, accessory Cas10 ssDNase and Cas7 ssRNase activities are alone sufficient to protect cells from phage infection (88, 92, 107), but in many cases inactivating mutations in the Cas10 ssDNase domain and in the Cas7 ssRNase active site do not affect phage replication, demonstrating that cOA signaling is the major mechanism of immune defense (43, 44, 92).

Role of Type III CRISPR Accessory Proteins

Type III CRISPR systems encode ancillary protein effectors that bind to cOA nucleotide immune signals and induce cell dormancy or cell death. Similar to CBASS Cap effectors, type III CRISPR ancillary proteins typically exhibit a modular architecture composed of either a CARF or a SAVED nucleotide-binding domain fused to a variable effector domain that provides immune protection (69, 103). Most effector domains in type III CRISPR function as nonspecific nucleases that induce cell growth arrest or cell death (103). Examples include Csm6 and Csx1, which degrade ssRNA (41, 101); Can2 and Card1, which degrade ssDNA (76, 89); and Can1 and NucC, which degrade dsDNA (75, 76). Notably, both SAVED domain-containing effectors and NucC DNase effectors are also encoded in CBASS antiphage defense operons (61, 68), highlighting that the modular nature of nucleotide immune signal effectors can allow exchange between multiple classes of antiphage defense. Bioinformatic analyses have identified additional ancillary proteins consisting of CARF or SAVED domains fused to nonnuclease effector domains (103). These include well-characterized examples like a SAVED–Lon protease fusion that cleaves a specific target protein to liberate an RNase toxin (90) and a SAVED–TIR fusion that depletes NAD⁺ (47). The functions of many ancillary proteins are undefined, highlighting a diversity of effector mechanisms that remain to be discovered in type III CRISPR immunity.

In addition to ancillary effector proteins, type III CRISPR systems often encode proteins that regulate cOA signaling and limit cell toxicity (6). The most widespread of these mechanisms is degradation of cOA, which is performed by three families of cellular enzymes named CRISPR-associated ring nucleases (Crn1, Crn2, and Crn3) (4, 5, 93). These proteins have unrelated folds and catalyze distinct mechanisms of cOA cleavage, suggesting that multiple nucleases have been independently co-opted throughout evolution to limit signaling in type III CRISPR immunity. The ring nuclease Crn1 consists solely of two CARF domains, demonstrating that this nucleotide-binding domain is also capable of degrading cOA (5). Interestingly, CARF domains of many ancillary effectors in type III CRISPR also have intrinsic cOA cleavage activity that autoterminates effector function and limits cellular toxicity (36, 53, 88, 100). Beyond degradation of cOA, direct cleavage of target ssRNAs by the Cas7 subunit is sufficient to deactivate the type III surveillance complex and provide an additional negative-feedback mechanism that limits Cas10 nucleotide immune signal generation (2, 41).

Phage Evasion of Type III CRISPR

Phages encode a diverse array of Acr proteins that inhibit nearly every aspect of CRISPR immunity (85, 115). AcrIII-1 is a phage immune evasion protein dedicated to inhibition of type III CRISPR that functions by rapidly degrading cOA to prevent activation of ancillary effectors (3) (**Figure 5c**).

First identified in the archaeal virus SIRV1 (*Sulfolobus islandicus* rod-shaped virus 1), AcrIII-1 is a small dimeric protein that is unrelated to the CARF domains found in cellular ring nucleases (3). Notably, viral AcrIII-1 cleaves cOA more than 100 times faster than cellular ring nucleases, resulting in an exceptionally fast turnover of cOA sufficient to outcompete Cas10 and enable immune evasion (2). Orthologs of AcrIII-1 are widely distributed among archaeal viruses, phages, and plasmids, suggesting that it is a highly effective form of immune evasion. Beyond AcrIII-1, the only other Acr protein that has been demonstrated to specifically inhibit type III systems is AcrIIB1. Interestingly, this protein was also identified in an archaeal virus that infects *Sulfolobus islandicus* and functions by directly binding the Cas10–Cas5–Cas7 surveillance complex (10), but the precise mechanism by which AcrIIB1 prevents immunity is not known (Figure 5c).

Eukaryotic Conservation

Although lacking clear direct structural homologs in eukaryotic cells, aspects of type III CRISPR immunity are shared in animal innate immunity. The Cas10 palm domain is distantly related to the nucleotidyltransferase domains of CBASS CD-NTase enzymes, human cGAS, and animal cGLR proteins that generate nucleotide immune signals in antiviral immunity (58). Following activation by cOA binding, the type III CRISPR effector proteins Csx1 and Csm6 use HEPN RNase domains to cleave target RNA (41, 101). In animal cells, the nucleotide immune signal 2′–5′-linked oligoadenylate, produced by oligoadenylate synthase enzymes, binds to an effector protein named RNase L that also uses HEPN domains to target RNA (16). Given the tremendous diversity of protein domains used in type III CRISPR systems and the known exchange of components with other antiphage defense systems, it is likely that additional connections with animal antiviral signaling pathways will be discovered.

CONCLUDING REMARKS AND OPEN QUESTIONS

Mechanistic analyses of nucleotide immune signaling continue to create new opportunities to explain general rules that control host–virus interactions. Features of CBASS, Pycsar, Thoeris, and type III CRISPR nucleotide immune signaling are conserved across all domains of life (98, 113), highlighting how this principle of immune defense plays a fundamental role in host–virus interactions throughout evolution. Notably, important questions in the field remain unanswered. First, how do diverse nucleotide immune signaling systems recognize phage infection? Exciting new studies have begun to identify phage-associated molecular patterns that activate CBASS immunity, but for many CBASS and all Pycsar and Thoeris systems the phage cues that activate host nucleotide immune signaling remain unknown. Second, what role does the bacterial host environment play in controlling abortive infection–based immune systems? Studies using ectopic overexpression of operons in model organisms like *E. coli* have been instrumental in uncovering new antiphage defense systems, but a key challenge remains to understand the function of these systems within native bacterial hosts. Emerging evidence supports the idea that many abortive infection systems in the native environment may trigger phage-induced cell dormancy instead of cell death (1, 29). Third, how do phages evolve to evade host immune defense? The use of closely related nucleotide signaling mechanisms allows for rapid exchange and diversification of components across distinct antiphage defense systems, but this mechanistic similarity is exploited by phages, as multiple viral evasion proteins can sequester or degrade nucleotide immune signals used in all four systems. It will be important to understand how nucleotide immune signaling systems shape the molecular arms race between phages and bacteria. Finally, what other connections exist between bacterial antiphage defense and animal antiviral immunity? The tremendous diversity of

bacterial antiphage defense systems and the phages they restrict represents a powerful resource to inform a new understanding of how both bacterial and animal cells control antiviral immunity.

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